TEST MORPHISMS AGAINST ARBITRARY FIRST ORDER THICKENINGS OF SPACES, NOT JUST AFFINE ONES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2765-2781 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2783-2789, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (only B has a complete proof).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Properties of Spaces, lemma cover-by-union-affines; descent/uniqueness of morphisms from surjective etale covers; definition-formally-unramified.

Source reference. Original lemma 02H8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2708–2713. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03FX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2789–2789. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. If $f : X \to Y$ is a formally unramified morphism of algebraic spaces over S, then given any solid commutative diagram*

$$\begin{array}{ccc} X & \longleftarrow & T \\ {\scriptstyle f}\downarrow & \nwarrow & \downarrow{\scriptstyle i} \\ Y & \longleftarrow & T' \end{array}$$

where $T \subset T'$ is a first order thickening of algebraic spaces over S there exists at most one dotted arrow making the diagram commute. In other words, in Definition 02H8 the condition that T be an affine scheme may be dropped.

Proof. This is true because there exists a surjective étale morphism $U' \to T'$ where U' is a disjoint union of affine schemes (see Properties of Spaces, Lemma 03FX) and a morphism $T' \to X$ is determined by its restriction to U'. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.